Lesson 1: Static Electricity

GCSE Physics: Electricity and Magnetism

Gemini Notebook

What subatomic particle carries a negative electric charge?

Hint: Orbits the nucleus.

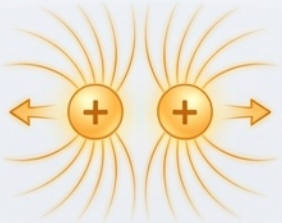
Answer: Electron.



What happens when two objects with the same charge are brought close?

Hint: Like charges...

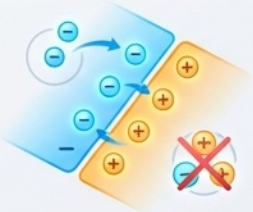
Answer: They repel.



Which subatomic particles are transferred when materials rub together?

Hint: Mobile outer particles.

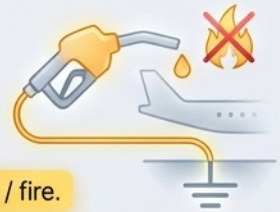
Answer: Electrons (never protons).



Why is a fuel pipe earthed before refuelling an aircraft?

Hint: Discharges static.

Answer: To prevent electrostatic spark / fire.



What is an electrical insulator? Give one example.

Hint: Holds charge locally.

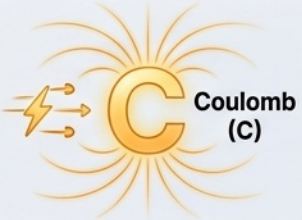
Answer: Material where electrons cannot flow freely (e.g. plastic / glass / polythene).



What is the unit of electric charge?

Hint: Symbol C.

Answer: Coulomb (C).



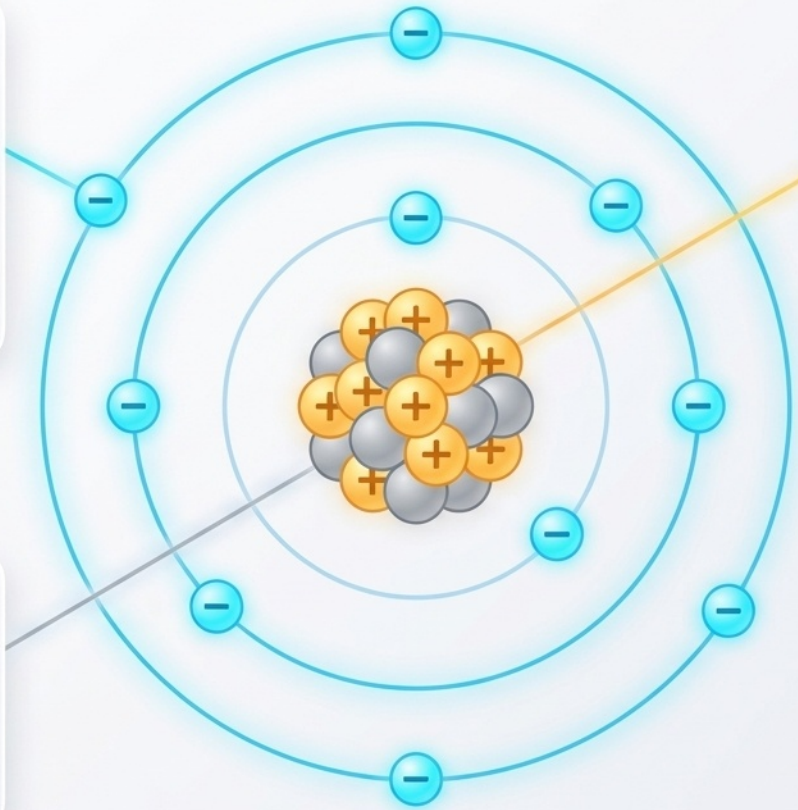
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Electrons

- Negatively charged particles.
- Orbit the nucleus at a distance.
- Outer electrons can be mobile.

Protons

- Positively charged particles.
- Locked tightly inside the central nucleus.
- Cannot move between atoms.

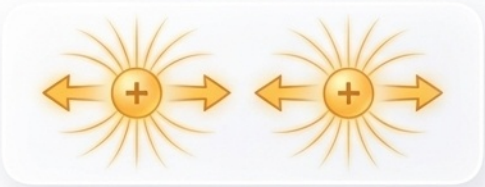


Neutrons

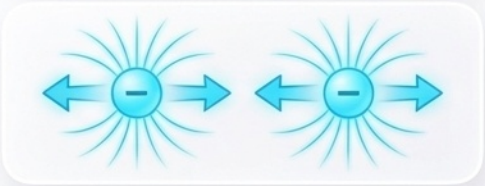
- Neutral particles (no electric charge).
- Locked inside the nucleus with protons.

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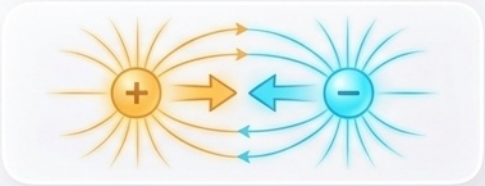
The Law of Electrostatics



Like charges repel



Like charges repel



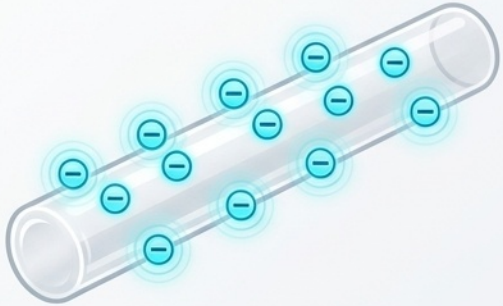
Opposite charges attract

Key Takeaway: Objects usually have zero net charge. Net charge only occurs when electron balances change.

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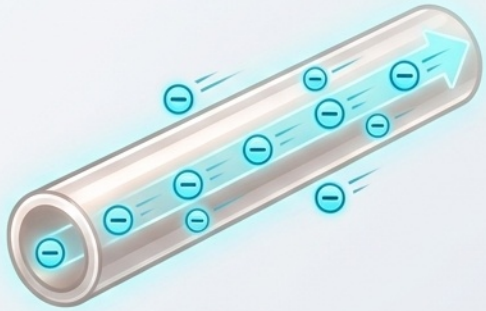
Why Static Charge Requires Insulators

Electrical Insulators



- Electrons cannot flow freely.
- Transferred charge remains locked in place.
- Allows static electricity to build up.

Electrical Conductors

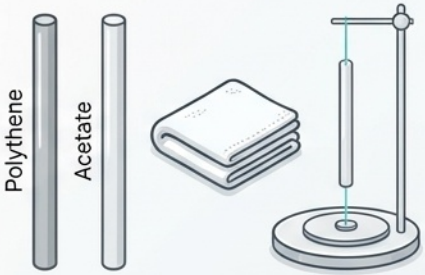


- Electrons flow freely through the material.
- Charge immediately dissipates.
- Cannot hold a static charge easily.

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Practical 1 Setup: Charging by Friction

Step 1: The Equipment



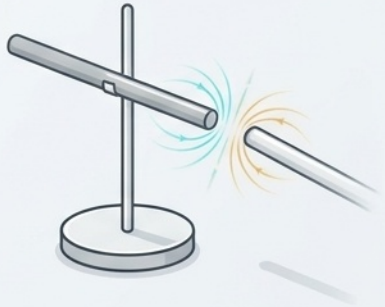
- Use insulating rods and a friction cloth.

Step 2: The Action



- Vigorously rub the rods with the dry cloth to cause friction.

Step 3: The Test

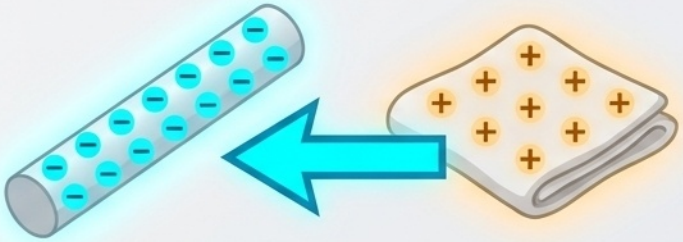


- Suspend one rod. Bring the second close to observe forces.

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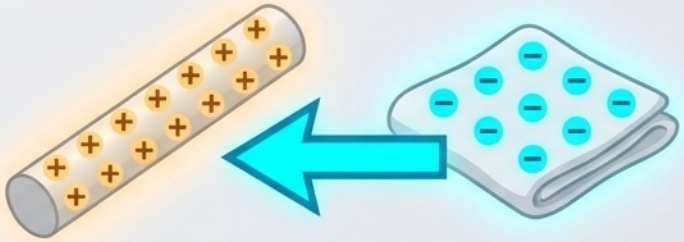
Practical 1 Results: Electron Transfer

Polythene Rod



- Gains electrons from the cloth.
- Becomes negatively charged.

Acetate Rod



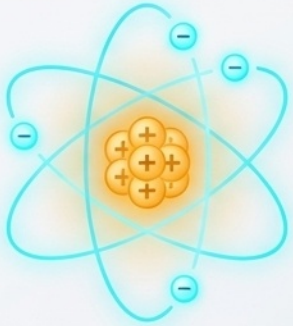
- Loses electrons to the cloth.
- Becomes positively charged.

Friction only moves electrons. The material that gains electrons becomes negative.

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Checkpoint: Pair & Share Challenge

When rubbing the acetate rod, why does it become positive by losing electrons rather than by gaining protons?

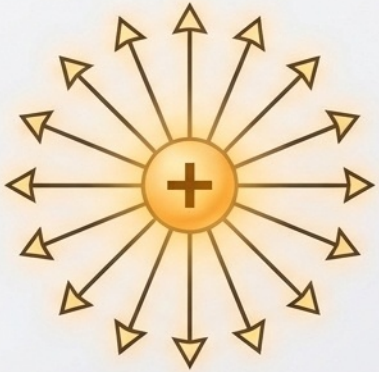


- Protons are locked inside the heavy central nucleus.
- Only outer electrons are mobile enough to transfer via friction.

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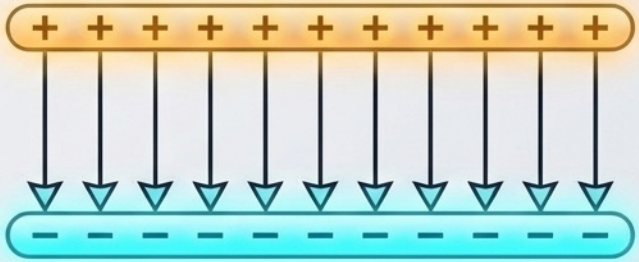
Mapping the Invisible: Electric Fields

Point Charges (Radial Field)



- Field lines extend radially around point charges.
- Arrows always point from positive to negative.

Parallel Plates (Uniform Field)

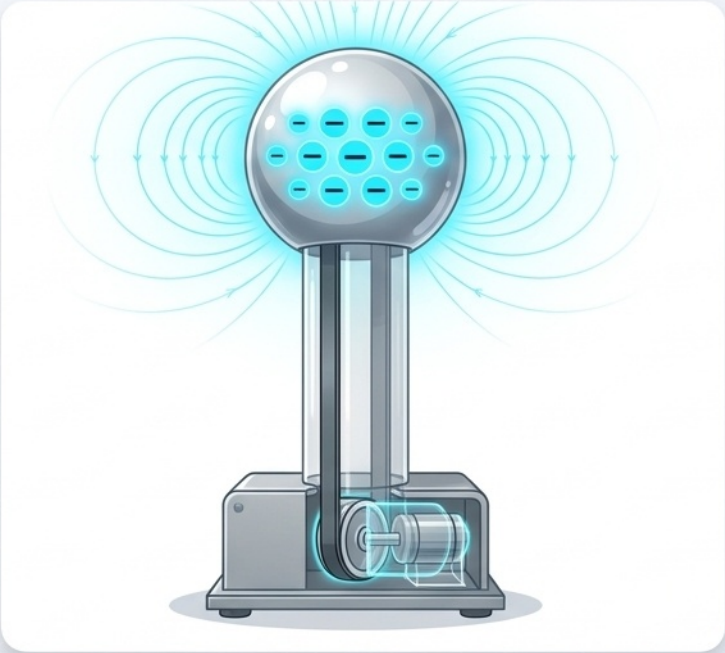


- Field lines are uniform and parallel between plates.
- Shows a constant force environment.

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Luminous Laboratory

Practical 2 Setup: Van de Graaff Generator



Protocol 1: Stand on an insulated rubber mat to prevent accidental earthing through the body.

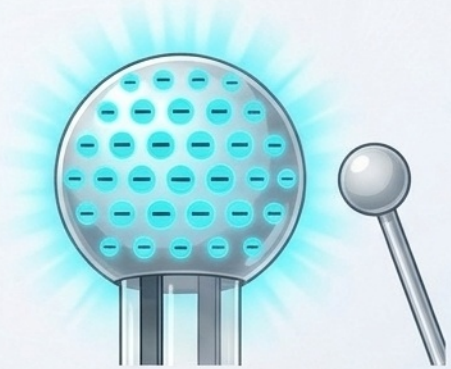
Protocol 2: Always use the grounded earthing wand to safely discharge the dome after use.

Protocol 3: Keep electronic devices (phones, pacemakers) far away from the intense electric field.

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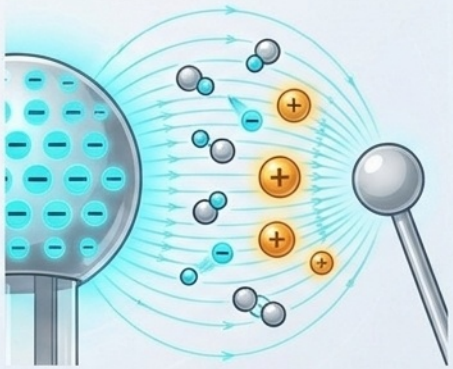
The Spark: Dielectric Breakdown of Air

Charge Accumulation



- Massive negative charge builds on the metal dome.

Intense Electric Field



- The electric field becomes so strong it strips electrons from air molecules (ionisation).

The Spark (Plasma Channel)



- The insulating air temporarily becomes a conductor. Electrons rush across, creating a visible spark.

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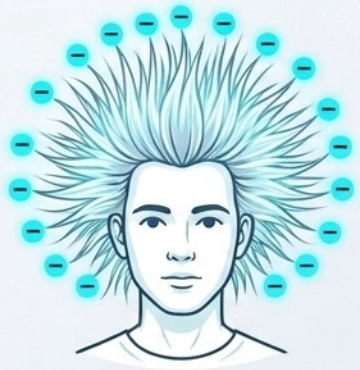
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Practical 2 Demonstrations

Hair Standing on End



- Individual hairs gain the same charge.
- Like charges repel, pushing hairs apart.

Flying Pie Tins



- Tins share the dome's negative charge.
- The intense repulsion overcomes gravity.

Gold Leaf Electroscope



- Charge flows down the stem.
- Stem and leaf gain the same charge and repel, showing deflection.

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▶ AUTHENTIC CCTV FOOTAGE | ELECTROSTATIC HAZARD | PETROLEUM EQUIPMENT INSTITUTE (PEI)

Stop Static: Refuelling Fire at Gas Pump

Investigating how friction between clothing and car upholstery generates thousands of volts of static charge, causing a spark discharge that ignites petrol vapour.



PHASE 1
Friction & Charge
Sliding on synthetic seat **transfers electrons**, charging driver up to ~20,000 V.

PHASE 2
Vapour Cloud
Petrol evaporates quickly, forming a **volatile flammable vapour** envelope.

PHASE 3
Spark Discharge
Touching metal nozzle triggers a **dielectric spark**, igniting the fumes.

PHASE 4
Earthing & Safety
Touch metal chassis (earthing) first. **Never pull out a burning nozzle!**

↗ Open video on YouTube (FPKen4QwY7I)

⚠ Safety Protocol: Never pull out burning nozzle • Back away & trigger Emergency Shut-off

Real-World Hazards & Earthing

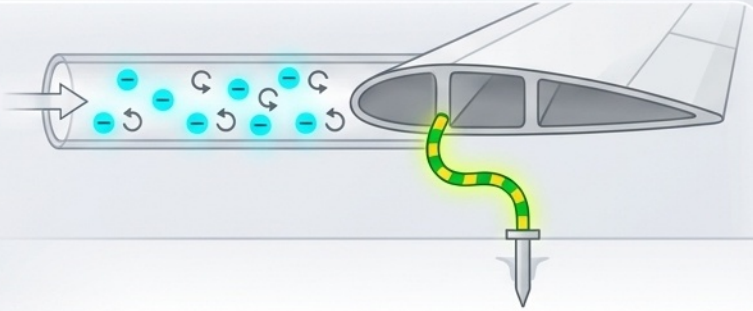
Hazard 1: Lightning (Nature's Spark)

- **Friction** in clouds builds massive charge, resulting in massive dielectric breakdown.



Hazard 2: Fuelling Aircraft & Cars

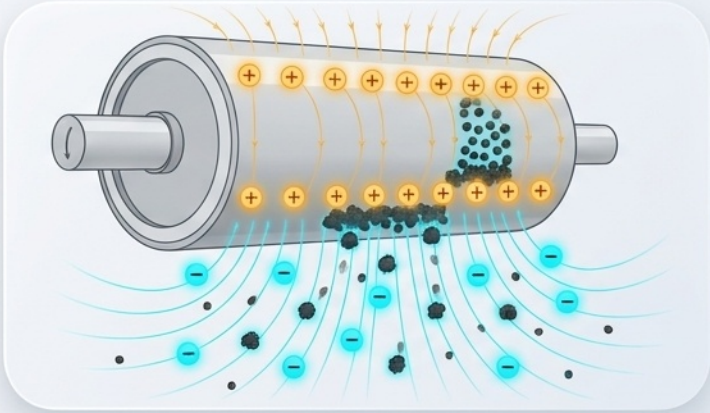
- **Friction** from fast-flowing fuel builds static charge.
- A spark could ignite fuel vapors.
- **Solution:** An earth wire provides a safe path for electrons to flow to the ground, preventing sparks.



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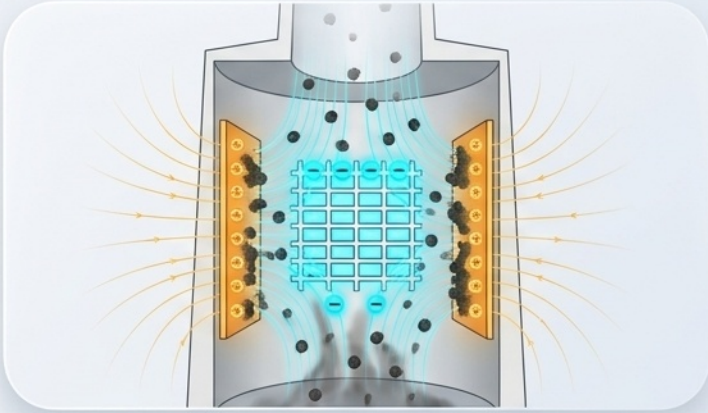
Harnessing the Charge: Real-World Uses

Photocopiers



- A positively charged image is projected onto a drum.
- Negatively charged toner powder is attracted to the image.
- The powder is transferred and melted onto paper.

Electrostatic Precipitators



- Fitted in factory chimneys to reduce pollution.
- Smoke particles are given a negative charge.
- They are attracted to positive collection plates and removed.

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Plenary: Quick-Fire Retrieval Race

Question: Which particle moves during friction: proton or electron?

Answer: Electron.

Question: A polythene rod gains electrons. What is its overall charge?

Answer: Negative.

Question: Why does a person's hair stand on end touching a Van de Graaff generator?

Answer: Like charges repel.

Question: Why do we connect an earth wire to an aircraft during refuelling?

Answer: To discharge static and prevent sparks.

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